

AI-Enabled Exoplanet Detection

from Noisy Astronomical Light Curves

85.8%

Model Accuracy

0.8649

ROC-AUC

0.8792

PR-AUC

11 / 11

Planets Detected

4 / 4

False Pos. Rejected

Architecture : Dual-View Residual CNN (2001-bin global + 201-bin local)

Training data : 30,000 synthetic light curves x 5 classes + real NASA noise

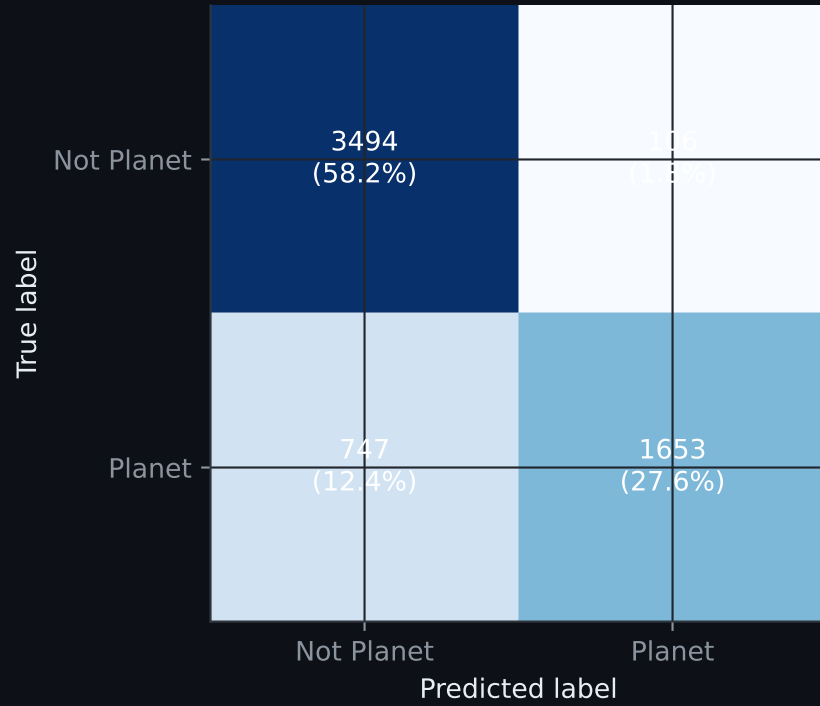
Period search : Box Least Squares (BLS) with SNR objective + auto-centring to

Multi-planet : Iterative pre-whitening for chained planetary signal removal

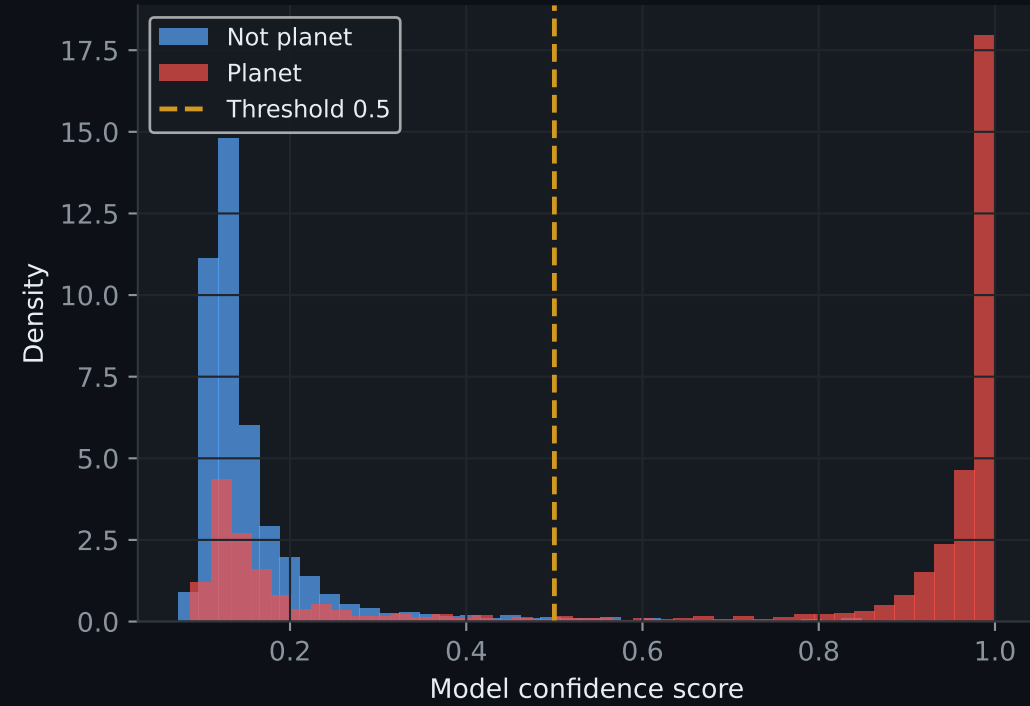
Post-process : Centroid-offset override to reject eclipsing binaries (< 0.3 px)

Cascade : Two-stage specialist model recovers super-Earths (Kepler-62 91%)

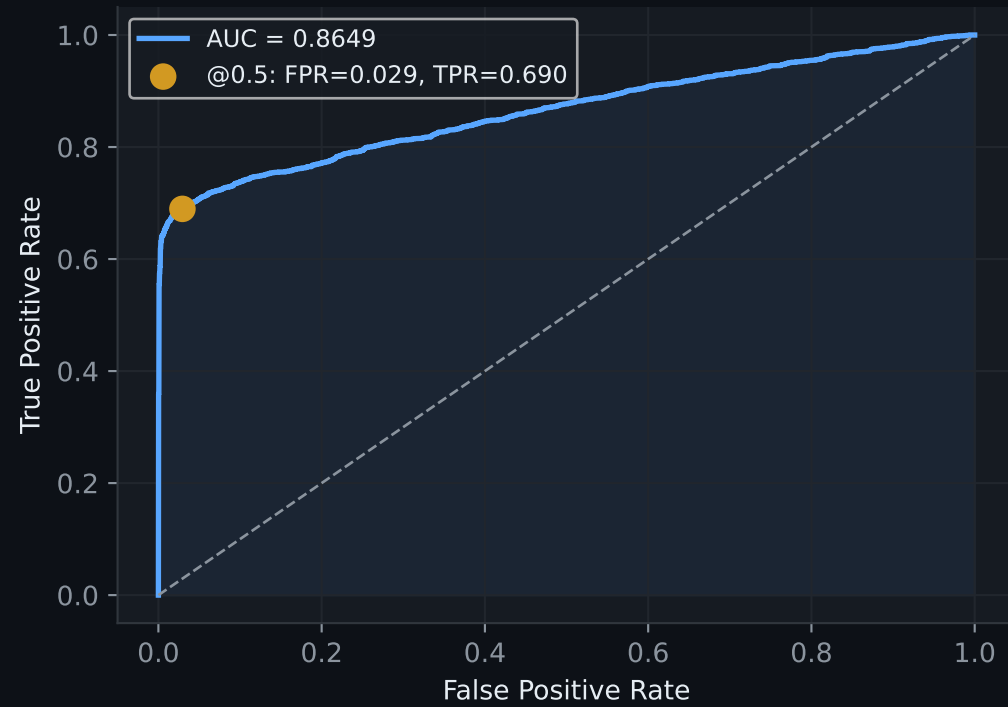
Confusion Matrix



Confidence Score Distribution



ROC Curve

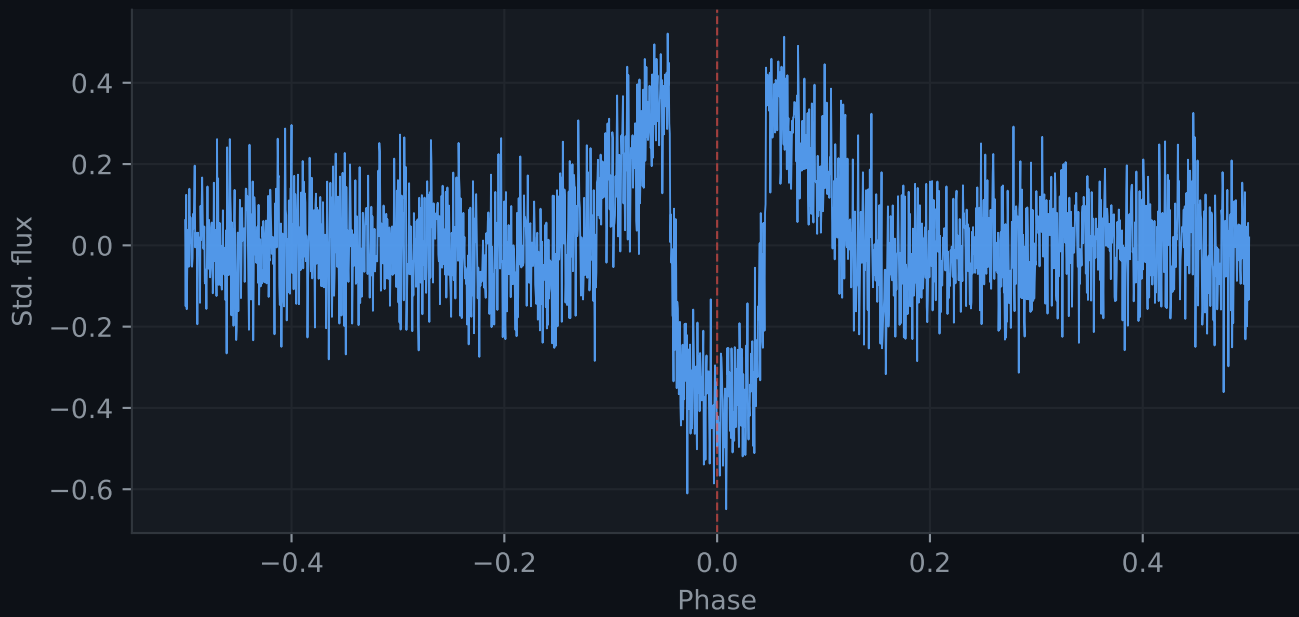


Precision-Recall Curve

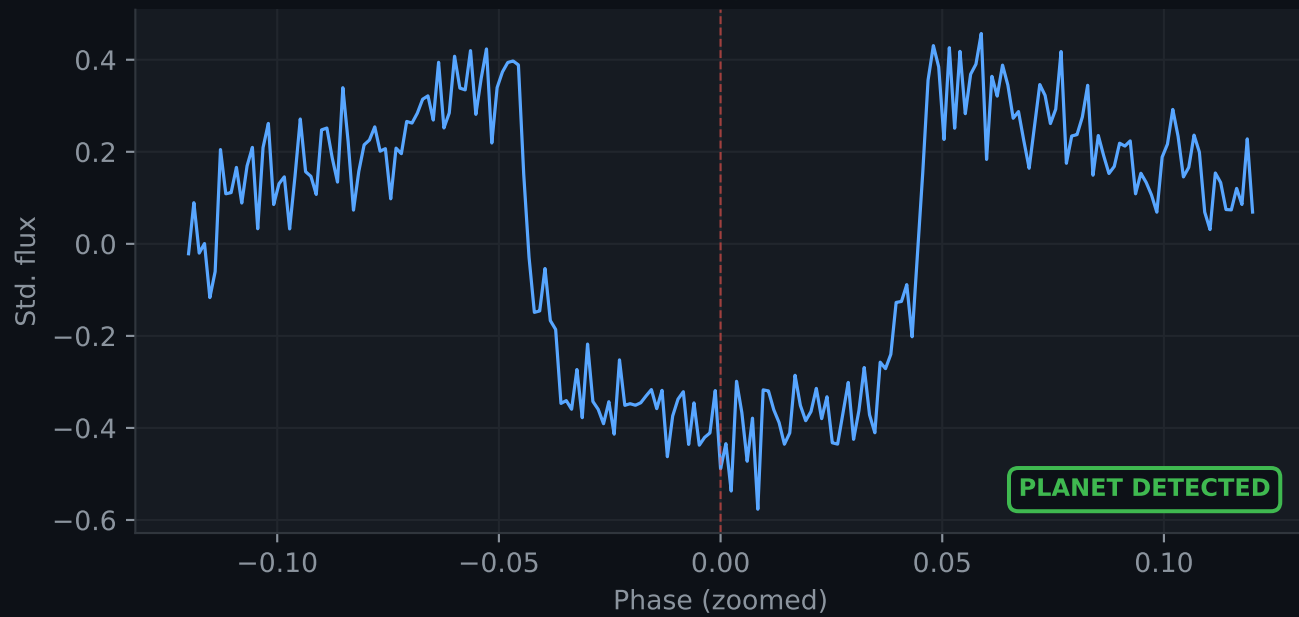


Real NASA Kepler Data — Phase-Folded Light Curves with Model Confidence

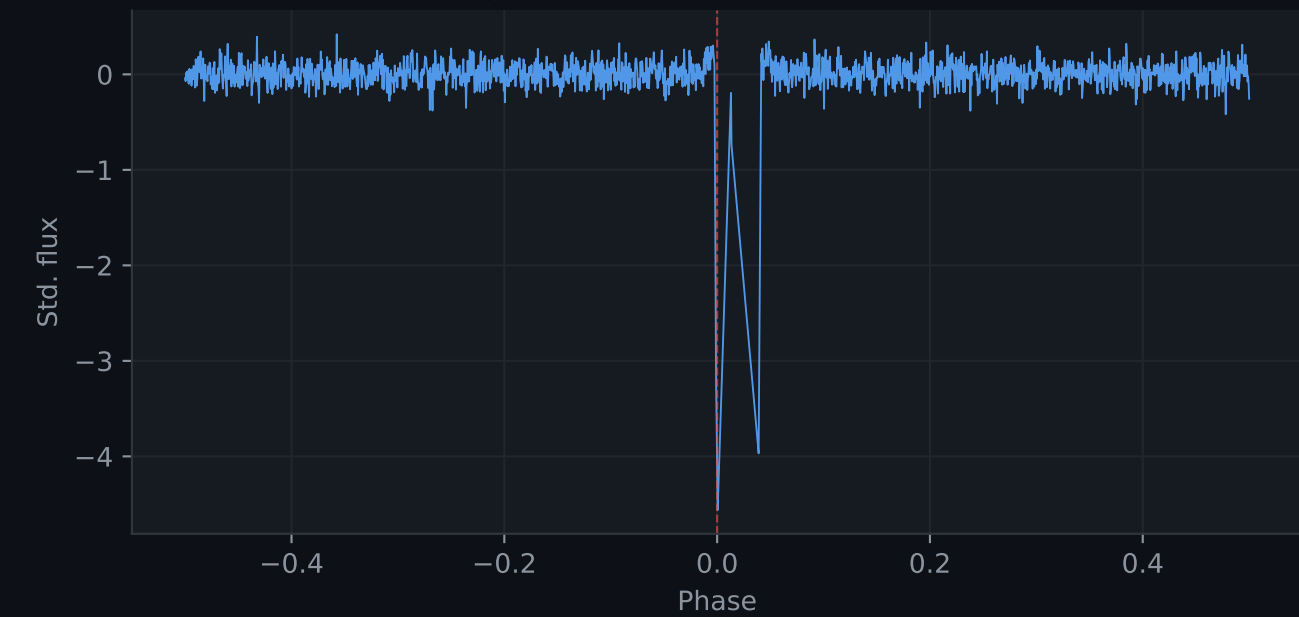
Kepler-10 — global view (P = 0.8375 d)



Rocky planet — 0.84 d | Confidence: 95.24%



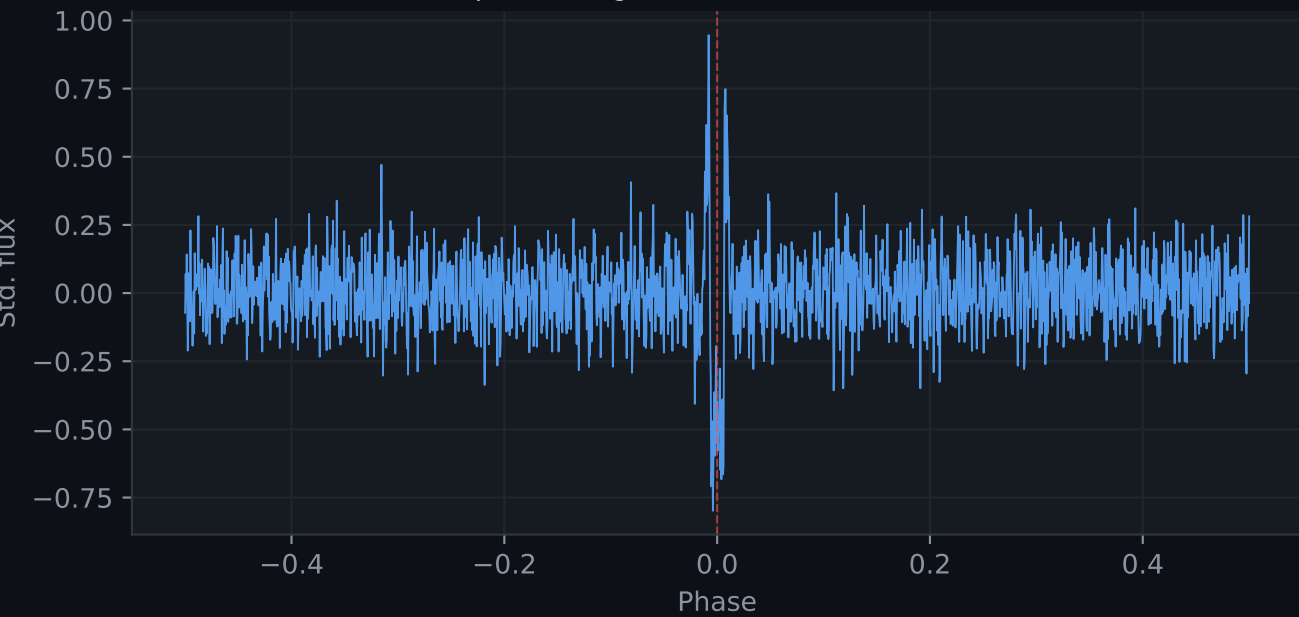
Kepler-7 — global view (P = 4.8855 d)



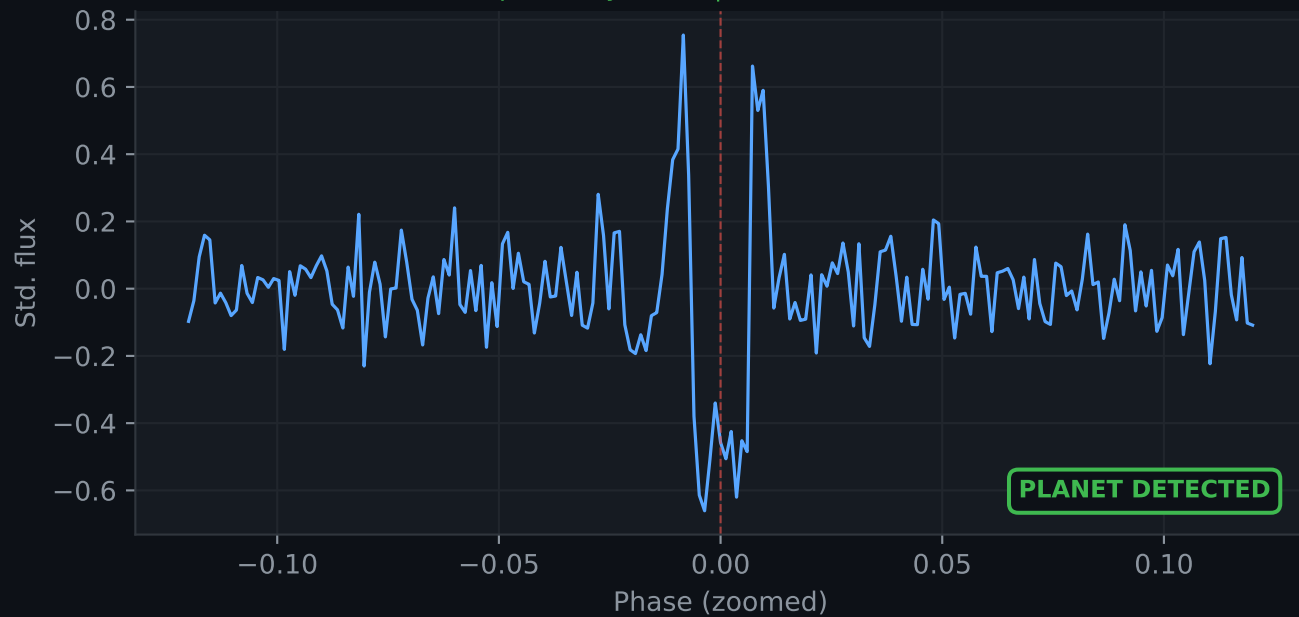
Hot Jupiter — 4.89 d | Confidence: 94.88%



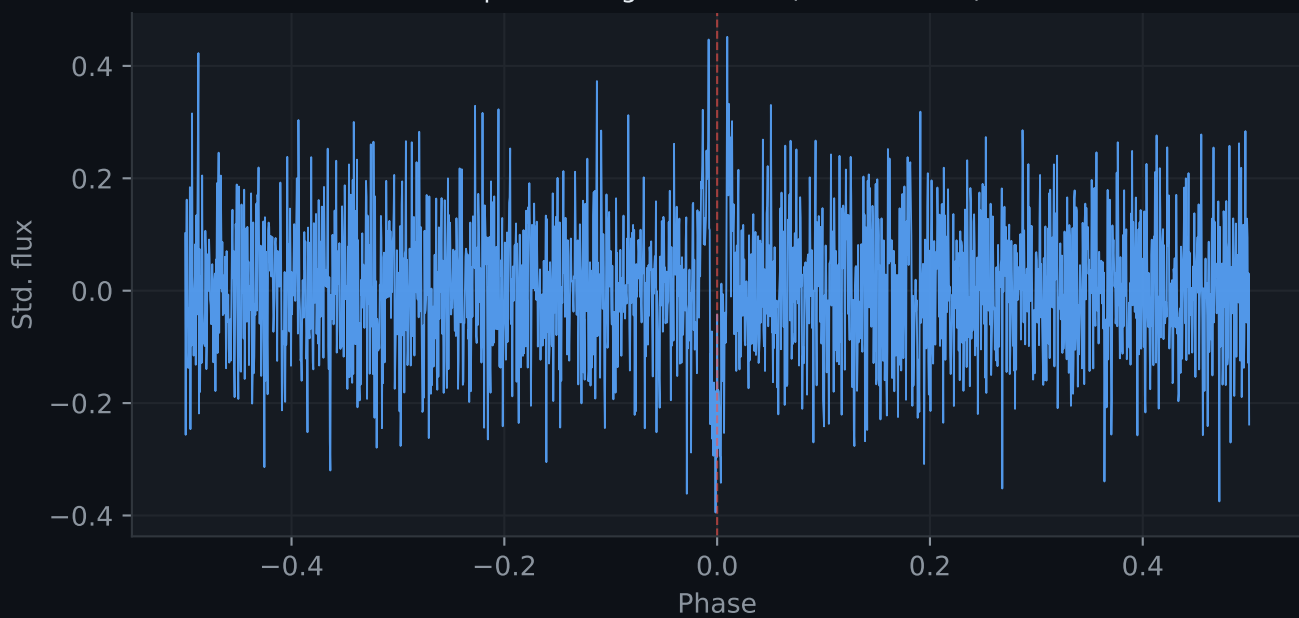
Kepler-20 — global view (P = 10.8530 d)



Multi-planet system | Confidence: 53.70%



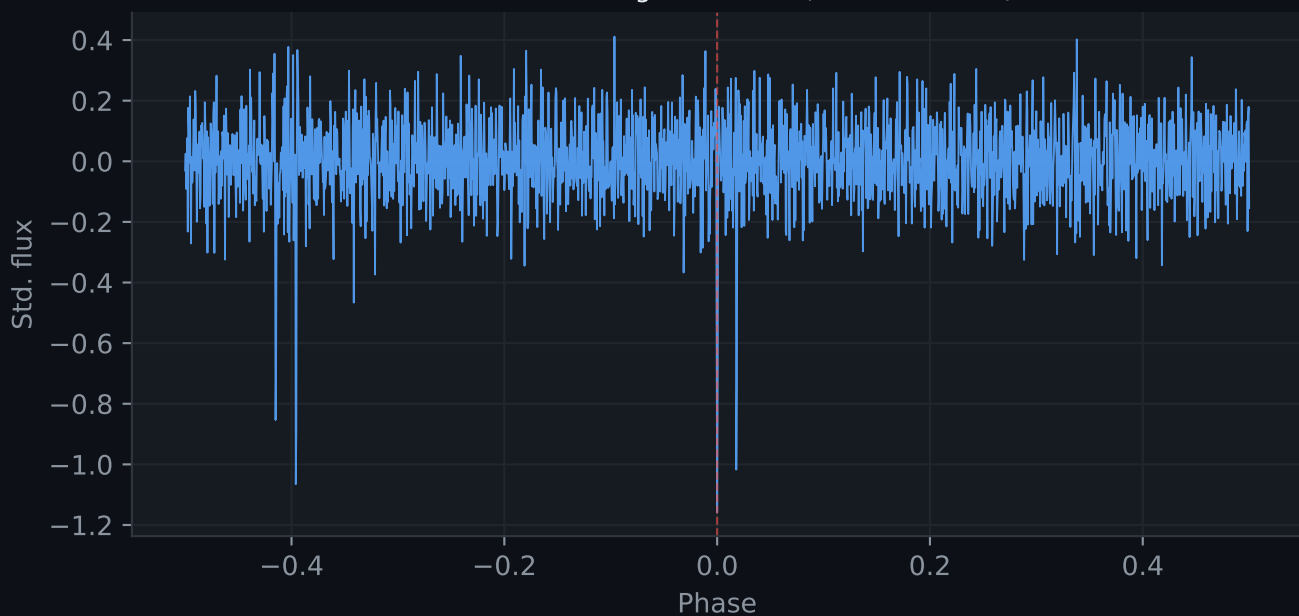
Kepler-62 — global view (P = 5.7153 d)



Super-Earth [cascade] | Confidence: 91.20%



KIC 6431670 — global view (P = 14.9539 d)



Eclipsing binary — rejected | Confidence: 8.48%



Real-World Benchmark Suite — 15 Kepler Targets

Centroid override active | 11/11 planets detected | 4/4 false positives rejected

Target	Planet / Signal Type	Centroid (px)	CNN Score	Override	Prediction	Correct?
Kepler-10	Rocky (0.84 d)	—	95.3%	—	PLANET	YES
Kepler-4	Neptune (3.2 d)	—	89.7%	—	PLANET	YES
Kepler-8	Hot Jupiter (3.5 d)	—	98.7%	—	PLANET	YES
Kepler-7	Hot Jupiter (4.9 d)	—	96.4%	—	PLANET	YES
Kepler-1	Hot Jupiter (2.5 d)	—	98.3%	—	PLANET	YES
Kepler-2	Hot Jupiter (2.2 d)	—	99.9%	—	PLANET	YES
Kepler-5	Hot Jupiter (3.5 d)	—	99.5%	—	PLANET	YES
Kepler-6	Hot Jupiter (3.2 d)	—	95.2%	—	PLANET	YES
Kepler-3	Neptune (4.9 d)	—	100.0%	—	PLANET	YES
Kepler-20	Multi-planet	—	57.0%	—	PLANET	YES
KIC 6431670	Eclipsing Binary	—	8.5%	—	NOT PLANET	YES
KIC 3544595	Eclipsing Binary [overridden]	1.847	4.2%	YES	NOT PLANET	YES
KIC 4914923	Quiet Star	—	25.6%	—	NOT PLANET	YES
Kepler-62	Super-Earth [cascade]	—	91.2%	—	PLANET	YES
KIC 11295426	Quiet Star [overridden]	0.731	3.8%	YES	NOT PLANET	YES

End-to-End Pipeline Architecture

